

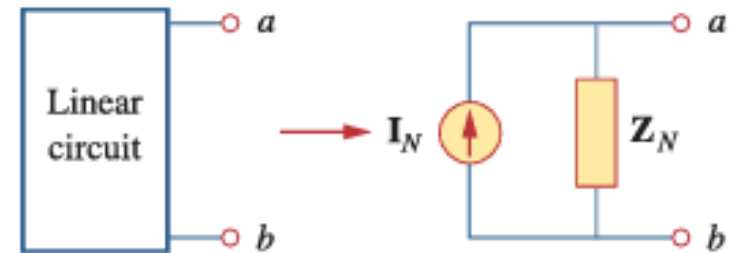
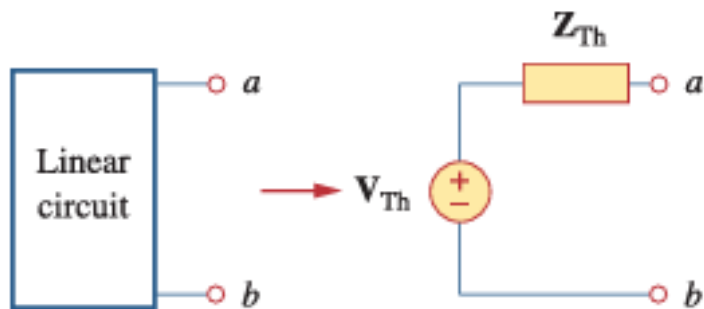
Module 3

Sinusoidal Steady-State Analysis

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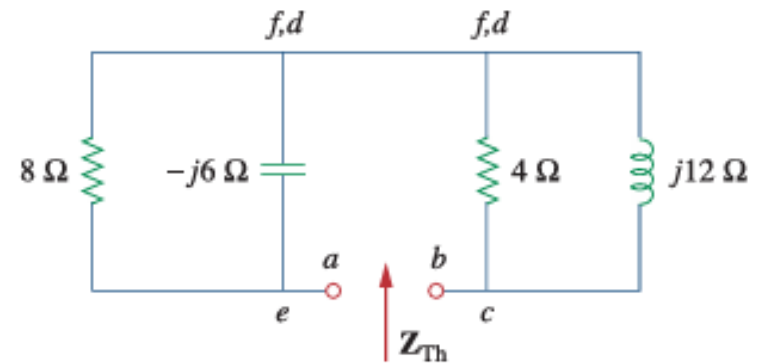
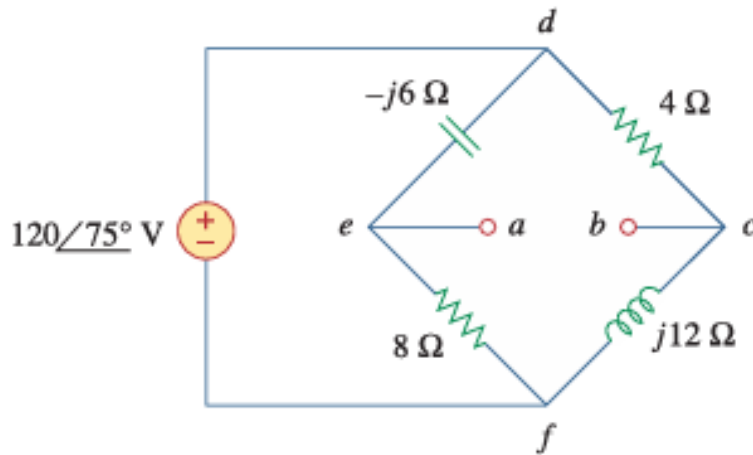
Thevenin's and Norton's Equivalent



$$V_{Th} = Z_N I_N, \quad Z_{Th} = Z_N$$

Thevenin's theorem

1. Obtain Thevenin's equivalent at terminals a-b of the circuit.



Find Z_{th} by setting the voltage source to zero:

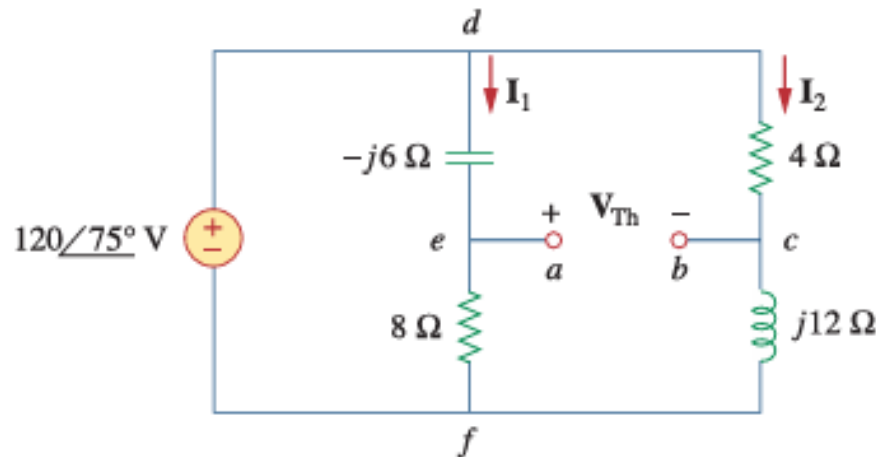
$$\mathbf{Z}_1 = -j6 \parallel 8 = \frac{-j6 \times 8}{8 - j6} = 2.88 - j3.84 \, \Omega$$

$$\mathbf{Z}_2 = 4 \parallel j12 = \frac{j12 \times 4}{4 + j12} = 3.6 + j1.2 \, \Omega$$

$$\mathbf{Z}_{Th} = \mathbf{Z}_1 + \mathbf{Z}_2 = 6.48 - j2.64 \, \Omega$$

Thevenin's theorem

Find V_{th} considering the circuit:



$$\mathbf{I}_1 = \frac{120\angle 75^\circ}{8 - j6} \text{ A}, \quad \mathbf{I}_2 = \frac{120\angle 75^\circ}{4 + j12} \text{ A}$$

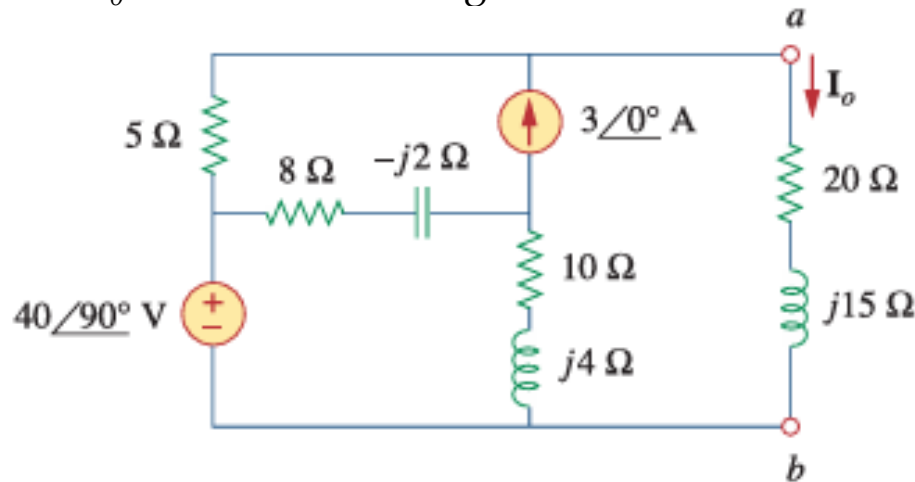
Applying KVL around loop $bcdeab$

$$\mathbf{V}_{Th} - 4\mathbf{I}_2 + (-j6)\mathbf{I}_1 = 0$$

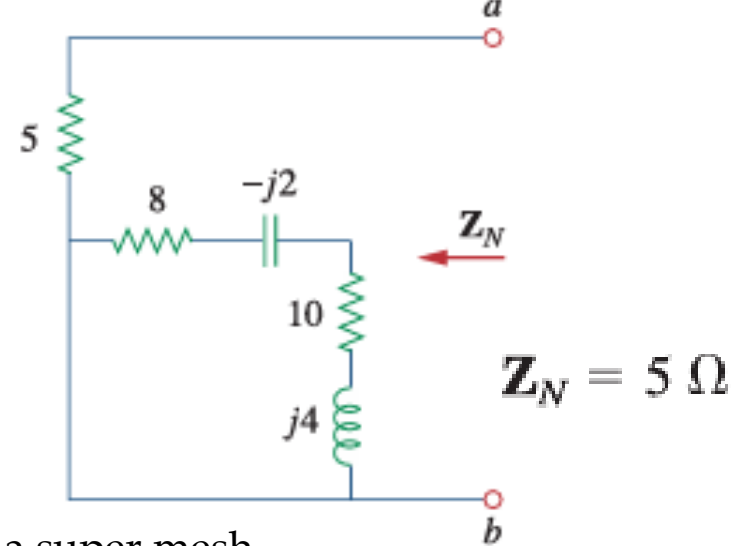
$$\begin{aligned} \mathbf{V}_{Th} = 4\mathbf{I}_2 + j6\mathbf{I}_1 &= \frac{480\angle 75^\circ}{4 + j12} + \frac{720\angle 75^\circ + 90^\circ}{8 - j6} \\ &= 37.95\angle 3.43^\circ + 72\angle 201.87^\circ \\ &= -28.936 - j24.55 = 37.95\angle 220.31^\circ \text{ V} \end{aligned}$$

Norton's theorem

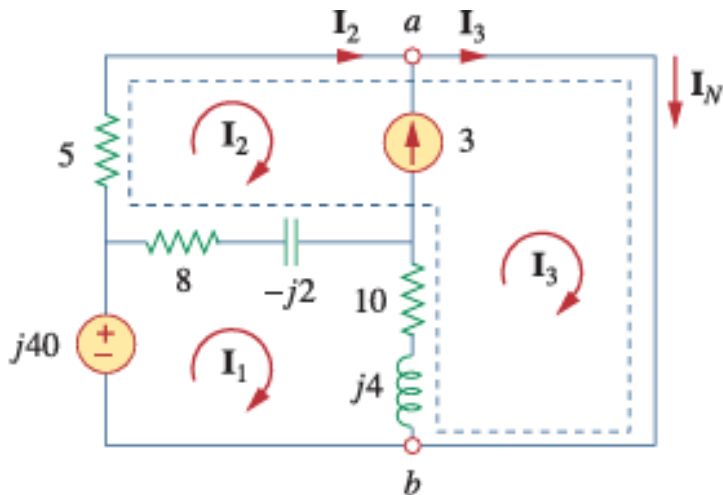
Find I_o in the circuit using Norton's theorem



Find Z_N considering the circuit



Find I_N considering the circuit: Mesh 2 and 3 form a super mesh.



For mesh 1

$$-j40 + (18 + j2)\mathbf{I}_1 - (8 - j2)\mathbf{I}_2 - (10 + j4)\mathbf{I}_3 = 0 \quad \text{--- 1}$$

For the supermesh

$$(13 - j2)\mathbf{I}_2 + (10 + j4)\mathbf{I}_3 - (18 + j2)\mathbf{I}_1 = 0 \quad \text{--- 2}$$

At 'a' due to the current source and current in mesh 1 and mesh 3:

$$\mathbf{I}_3 = \mathbf{I}_2 + 3$$

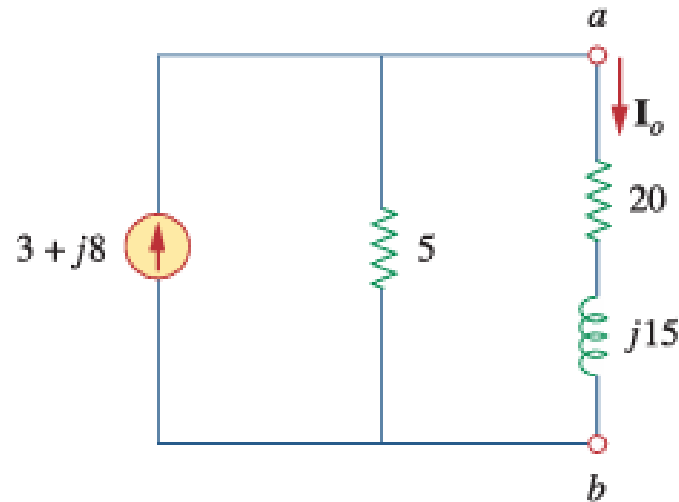
Norton's theorem

Add eqn.1 and eqn.2

$$-j40 + 5\mathbf{I}_2 = 0 \quad \Rightarrow \quad \mathbf{I}_2 = j8$$

$$\mathbf{I}_3 = \mathbf{I}_2 + 3 = 3 + j8$$

$$\mathbf{I}_N = \mathbf{I}_3 = (3 + j8) \text{ A}$$



$$\mathbf{I}_o = \frac{5}{5 + 20 + j15} \mathbf{I}_N = \frac{3 + j8}{5 + j3} = 1.465 \angle 38.48^\circ \text{ A}$$

Next class: Maximum Power Transfer Theorem